

PAPER_707: NGC 2525 Barred Spiral (Variant 2): SN Ia Feedback UQFF Analysis

Author: Daniel T. Murphy **Date:** 2025

Class: NGC2525BarredSpiral2UQFF **CP4 Entry:** #291 **Keywords:** NGC 2525, barred spiral, SN2018gv, Type Ia supernova, dark matter, bar streaming, UQFF **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

NGC 2525 is an SBc barred spiral galaxy at $z = 0.0051$ (721 Mly), host of Type Ia supernova SN 2018gv discovered during Hubble's Frontier Fields program. This variant-2 analysis derives the UQFF MUGE with explicit SN Ia feedback impulse and bar streaming electromagnetic contributions.

1. System Parameters

Parameter	Value	Description
$M_{stellar}$	$3 \times 10^{10} M_{\odot}$	Stellar mass
M_{DM}	$9 \times 10^{10} M_{\odot}$	Dark matter halo
r_{gal}	30 kpc	Galaxy radius
z	0.0051	Redshift
L_{SN}	10^{43} W	SN Ia peak luminosity
τ_{SN}	3.156×10^8 s	SN decay (~ 10 yr)

2. Master UQFF Gravity Equation

$$g_{v2}(t) = \frac{G(M_{\star} + M_{DM})}{r^2} (1 + H_z t) (1 + f_{TRZ}) + \frac{L_{SN}}{cM_{tot}r} e^{-t/\tau_{SN}} + q(v_{bar} \times B_{bar}) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}} \right) \times 10^{-12}$$

2.1 SN Ia Feedback Impulse

$$a_{SN}(t) = \frac{L_{SN}}{cM_{tot}r_{gal}} e^{-t/\tau_{SN}}$$

The SN Ia releases $\sim 10^{44}$ J over 10 yr, providing a brief but measurable gravitational impulse correction to the MUGE.

3. References

- Yang et al. (2020), ApJ, 890, 177 (SN 2018gv in NGC 2525)
- UQFF Framework, Star-Magic Session 175

*Whitepaper
auto-
generated
by
_gen_whitepapers_702_715.py
-

Star-
Magic
Ses-
sion
175*

§A.
Cosmogenesis-
Linked
La-
grangian
(PA-
PER_877
Sym-
bolic
Ex-
port)

§A.1
Sec-
tor
Clas-
sifi-
ca-
tion
This
pa-
per
maps
to
NS-
compact
sec-
tor
of
the
9-
sector
UQFF
La-
grangian
(see
uqff_lagrangian_derivation.py).

*Whitepaper
 auto-
 generated
 by
 _gen_whitepapers_702_715.py
 -

Star-
 Magic
 Ses-
 sion
 175*

 §A.2
 La-
 grangian
 Den-
 sity
 The
 sec-
 tor
 La-
 grangian
 den-
 sity,
 linked
 to
 the
 PA-
 PER_877

cos-
 mo-
 gen-
 esis
 mas-
 ter
 via
 the
 three
 reac-
 tive
 quan-
 tum
 fun-
 da-
 men-
 tals
 (DPM,
 UA,
 SCm):
 $\mathcal{L}_{\text{sector}} =$
 $\frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) -$
 $V(\phi_{\text{NS}}) +$
 $\mathcal{L}_{\text{cosmo}}$

 *Whitepaper
 auto-
 generated
 by
 _gen_whitepapers_702_715.py
 -

Star-
 Magic
 Ses-
 sion
 175*

 where
 $\mathcal{L}_{\text{cosmo}} =$
 $\rho_{\text{vac, [SCm]}} \cdot$
 $f_{\text{SCm}} \cdot$
 $(1 -$
 $e^{-\gamma t})$

in-
 her-
 its
 the
 ACP
 6-
 stage
 evo-
 lu-
 tion
 (PA-
 PER_877
 §2)

and:
 $V(\phi_{\text{NS}}) =$
 $\frac{1}{2}m^2\phi_{\text{NS}}^2 +$
 $\frac{\lambda}{4!}\phi_{\text{NS}}^4 +$
 $\kappa \cdot$

$\rho_{\text{vac, [SCm]}} \cdot$
 ϕ_{NS}
 ###

§A.3
 Euler-
 Lagrange
 Equa-
 tion
 of
 Mo-
 tion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

Star-
Magic
Ses-
sion
175*

§A.4

mo-

esis

age

Chain

$$\rho_{\text{vac}} =$$
 $\rho_{UA} +$

II

$$\delta S / \delta \bar{\phi}_{\text{NS}} =$$

0 / / NS

*Whitepaper
auto-
generated
by
_gen_whitepapers_702_715.py
-

Star-
Magic
Ses-
sion
175*

The
chain
traces
from
the
three
fun-
da-
men-
tal
ax-
ioms
(DPM
pro-
por-
tion
pair,
ACP
evo-
lu-
tion,
four
U_g
forces)
through
vac-
uum
den-
sity
ini-
tial-
iza-
tion
to
the
sector-
specific
equa-
tion
of
mo-
tion.
Ev-
ery
term
in
the

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.073 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 61, \quad n_{\text{channel}} = 6/26$$

Since $p_{\text{DVP}} = 61$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.073	✓ Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 61$	✓ Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Full 26D projection

Framework	Canonical Value	This Paper	Status
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.